NRSG 224 Patho – Class 8 Study Guide (Circulatory 3 / Respiratory 1), FILLED

*Filled answers are in **bold**. Sources: "Slide N" = the recorded lecture deck. "@h:mm:ss" = timestamp in the recorded lecture audio. Quotes are the professor's words. Anything not from her lecture or slides is in the "Not from the professor" section at the end.*

Circulatory Function, Part 3

Selected commonly-occurring alterations in cardiac rhythm

I. Electrocardiogram (ECG/EKG): recording of the heart's electrical activity. Dysrhythmias change the patterns of electrical activity, and those changes show up in the EKG tracing. - A. P wave: depolarization of SA node and atria - B. QRS complex: ventricular depolarization - C. T wave: ventricular repolarization - D. Atrial repolarization happens during ventricular depolarization, so it is hidden in the QRS complex - E. "Normal" means "normal sinus rhythm": a P wave before each QRS complex, and R-R intervals relatively constant over time *(she adds: "the R's have to be evenly spaced apart. Because that indicates that the rhythm is regular", Slides 19–20, @0:15:39)*

Conduction system (table)

Normal physiology. Function: generate and transmit action potentials that trigger heart-muscle contraction, and coordinate atrial and ventricular activity. - SA node: 60–100/min. AV node: 40–60/min, delays the impulse *("to allow time for the atria to finish contracting, emptying its blood into the ventricles so that the ventricles are nice and full", Slide 6)*. - Purkinje system: conduction is rapid, so all of both ventricles is excited almost at once → ↑ **CO** *("so we get a nice, strong force of contraction, we get good stroke volume, we get maximum cardiac output", Slide 8)*. If the Purkinje fibers have to pace the heart, the rate is 15–40/min. - Membrane potentials, automaticity, absolute and relative refractory periods, and the supernormal period are already printed on the handout (Slides 9–15).

Pathological change: changes in conduction or rhythm that can interfere with pumping. R/t altered automaticity, altered conductivity, and reentry. PF are printed on the handout.

Nursing problem: Decreased cardiac output. *"the main reason why we care whether people have dysrhythmias is that it can drop their cardiac output"* (Slide 21, @0:17:06). She explains each mechanism: - **Automaticity:** "If your rate is too slow, cardiac output is going to drop. If your rate is too fast, the heart doesn't have time to fill in between each beat." With an ectopic pacemaker, the ventricles may contract before the atria, so ↓ **ventricular filling** → ↓ **CO** (Slides 22–26, @0:19:56–0:27:11). - **Conductivity:** delays or blocks from scar tissue (old MI), electrolyte imbalance, or inflammation make "different parts of the ventricle wall … depolarize at different times." You lose the strong contraction, so ↓ **SV and** ↓ **CO** (Slides 27–28, @0:27:11). - **Reentry:** "an impulse that reenters an area of the heart wall that is already depolarized and depolarizes it again". This disrupts atrial/ventricular timing and ventricular filling, so ↓ **CO** (Slides 29–33, @0:28:39–0:33:34).

Example of disorder: cardiac dysrhythmias.

Assessment findings: *(Slides 35–36, @0:36:44–0:38:56)* - **Assessment findings depend on which dysrhythmia the patient has.** - **Rate too slow** → ↓ **CO. Rate too fast** → ↓ **CO** (no time to fill). A fast rate also means ↑ **cardiac workload and** ↑ **O2 demand.** With coronary atherosclerosis, this "might actually be setting this person up for a heart attack." - **Signs and symptoms of** ↓ **CO, by severity:** - **Mild:** tired, poor activity tolerance - **More severe:** dizziness, fainting, hypotension - **Really bad:** heart failure or even shock - **A-fib: irregular radial pulse.** "you would have to count this person's pulse for a full minute." There is **no distinct P wave** ("scribbly stuff"). Watch for **clots** → **emboli** → **ischemic stroke** ("one of the most common causes of ischemic strokes") (@0:43:04).

Interventions: *(Slides 51–56, @0:57:09–1:02:36)* - **First, find and remove the cause if you can**, e.g., "correct the electrolyte imbalance and hopefully the dysrhythmias should go away." - **Antidysrhythmic drugs:** "most of these anti-dysrhythmic drugs work by changing the permeability of the heart muscle cell membrane to different electrolytes" (potassium channel blockers, calcium channel blockers, etc.). That "changes automaticity, … affects conductivity, … changes the refractory periods, and therefore changes the generation of action potentials." - **Electrical interventions:** - **Electronic pacemaker:** for certain AV blocks and brady-dysrhythmias. It has a generator plus leads in the atria and ventricles. It senses the heart's rate, and "if the heart rate drops below a certain level, then the pacemaker will generate an electrical impulse that should stimulate the heart to contract." You can see a pacemaker-sized lump under the skin of the chest wall. - **Defibrillation:** "a much bigger electrical stimulus … to hopefully depolarize all the heart muscle cells all at the same time … interrupt all the disorganized impulses … and reset the heart so that the SA node can hopefully regain control." It only works on **shockable** rhythms (the AED decides). Sometimes medications can change a non-shockable rhythm into a shockable one.

II. Common examples (she said she will not test rhythm-strip identification, @0:38:56)

- **A. Sinus node dysrhythmias** (Slide 37, @0:40:27). Sinus brady < 60 and sinus tachy > 100. The EKG pattern is otherwise normal. **Either can be normal:** tachy during exercise, brady in very fit athletes.
- **B. Atrial fibrillation** (Slide 38, @0:43:04). The atria quiver, so ↓ **ventricular filling** → ↓ **CO**. Pooling leads to thrombus and systemic emboli (stroke).
- **C. AV block** (Slides 39–41, @0:47:18–0:51:09):
- 1st degree: every P has a QRS, but the P-to-QRS space is longer
- 2nd degree: some P waves have no QRS after them
- 3rd degree: P waves and QRS complexes are each evenly spaced but unrelated, and a ventricular ectopic pacemaker takes over
- **D. Ventricular dysrhythmias** (more serious, because the ventricles are the main pumps):
- **PVC:** ventricular filling may be ↓. The complex is wide and bizarre and comes early. **Atrial kick** (Slide 47, @0:53:03): *"when the atria have a chance to have a nice strong contraction and fully empty into the ventricles and give us maximal cardiac output."* A PVC "probably happens at the same time that the atria are contracting," so **atrial kick is lost**. An occasional PVC with no heart disease probably doesn't matter. Frequent PVCs predispose to more serious dysrhythmias.
- **V-tach:** 100–250/min → ↓ diastolic filling time → ↓ **CO**. "there's totally not going to be any atrial kick". The tracing looks like PVCs strung together (@0:54:49).
- **V-fib:** the ventricles quiver but don't contract → ↓ **CO** (it "totally disappear[s]"). EKG tracing: **totally disorganized, nothing orderly.** "unless you can get this patient out of this rhythm quickly, they're going to die" (@0:55:55).

Respiratory System, Part 1

Function: gas exchange. **Pathological change:** gas exchange impaired. **Nursing problem: Impaired gas exchange, which leads to hypoxia (not enough O2) and hypercapnia (too much CO2).** "if you don't have enough oxygen, cells can't make ATP, they're going to malfunction and die" (Slide 59, @1:02:49).

Conducting airways

They warm, moisten, and filter air. "Gas exchange does not happen in the conducting airways" (@1:03:41). - **Outer layer:** cartilage rings keep the airway from collapsing on inspiration. - **Middle layer, smooth muscle:** "the role of the smooth muscle is to control the diameter of the airways." - Patho change: bronchospasm. - **Nursing problem: Ineffective breathing / impaired airflow (airway obstruction).** "The airways get so constricted we can't move much air through. It [would] almost be like an airway obstruction. So even though gas exchange doesn't [happen] in the conducting airways, [they] can still cause respiratory problems" (@~1:06:30, re-transcribed). - **Inner layer, mucociliary blanket ("escalator")** (Slide 63): the mucus traps dust and particles, and the cilia sweep it up to the pharynx to be coughed up or swallowed. - Patho change: impaired function (smoking "kind of paralyzes the little cilia"). - **Nursing problem: Risk for infection** ("pathogens that are trapped in the mucus don't get swept out") and **airway inflammation** ("some irritant that was trapped in the mucus that didn't get moved on out") (@1:05:30). - **Swelling of mucosa:** - **Nursing problem: Narrowed airway → impaired airflow.** "It might cause some narrowing of the airway, so it might kind of impair airflow" (@~1:06:25, re-transcribed).

Respiratory tissues (alveoli and pulmonary capillaries)

- **Type I cells** (site of gas exchange):
- **Nursing problem: Impaired gas exchange.** "If something goes wrong with your type 1 alveolar cells, you're gonna have a problem with gas exchange" (@1:14:13).
- **Type II cells** produce surfactant → ↓ surface tension in the alveoli. Surface tension in the fluid film "starts pulling the alveoli shut."
- Patho change: impairment of Type II cells → ↓ surfactant.
- PF: ↓ deep breathing / chest expansion; toxic gases, smoke, anesthesia; O2 toxicity.
- **Nursing problem: Atelectasis** (Slide 72): collapse of alveoli. It may be a small group or a whole lobe. Either way, "you're not going to have good gas exchange." It is a complication of immobility: lying supine → shallow breathing → no surfactant (@1:11:07).
- **Assessment findings: decreased breath sounds over the area of collapse.**
- **Interventions** (Slide 73):
    - **TCDB (turn, cough, and deep breathing exercises)**
    - **Incentive spirometer:** the patient *inhales* (doesn't blow) as deep as possible, "like 10 times every hour while they're awake." Deep breaths stimulate surfactant (@1:13:13).
- **Alveolar macrophages:**
- **Nursing problem: Infection and inflammation** "from foreign stuff that didn't get … cleaned up" (e.g., smoking kills them) (@1:14:13).

Pleura

Function: ↓ friction between chest wall and lung. The negative-pressure "vacuum" keeps the elastic lung from recoiling (@1:15:33). - **Inflammation of the pleura (pleuritis/pleurisy):** - **Nursing problem: Pain: pleuritic chest pain** (Slide 79, @1:18:44). It is "a sharp, stabbing type of chest pain that is aggravated by respiratory movement" (coughing or deep breaths). It is **often unilateral.** - **vs. cardiac chest pain:** tightness, "elephant sitting on my chest" - **vs. musculoskeletal pain from prolonged coughing:** bilateral and not sharp or stabbing - **Air enters the pleural space (pneumothorax):** - **Nursing problem: Lung collapse → impaired ventilation/gas exchange.** "there's nothing to help keep that lung expanded, so it's going to recoil and collapse." - **Intervention: chest tubes** "to kind of suck all that air out … so that the lung can re-expand" (detailed in Adult Health 3 / critical care) (Slides 77–78, @1:18:08).

I. Mechanics of breathing (Slides 80–90, @1:21:10–1:34:15)

A. Respiratory pressures - a. Inspiration: chest cavity size ↑ (respiratory muscles contract) → pressure falls below atmospheric → air moves in. - b. Expiration: chest cavity size ↓ → pressure rises above atmospheric → air moves out. - Respiratory muscles: - a. Muscles of inspiration work to ↑ chest cavity size. - b. Expiration (↓ chest cavity size) is mostly passive. - She adds that if the respiratory muscles can't contract (e.g., spinal cord injury), "air is not going to move in and out anymore."

B. Lung compliance. It takes ↑ pressure to move the same amount of air into a noncompliant lung (balloon analogy, @1:24:56). 1. **Things that might ↓ compliance** (@1:24:56–1:27:02): - **Scar tissue / fibrosis** ("not as elastic as lung tissue") - **Pulmonary edema** (fluid in the lungs is heavy) - **Lack of surfactant** (you can't overcome surface tension) 2. **Things that might ↑ compliance** (@1:27:02): - **Diseases that cause lung tissue to lose its elastic recoil.** The lung is "easy to inflate, but … hard to deflate," so old deoxygenated air can't get out and fresh air can't get in. "You don't want lungs that are non-compliant. You don't want lungs that are too compliant."

C. Airway resistance. ↑ airway resistance → ↑ respiratory effort. - It depends mostly on airway diameter (boba straw vs. coffee stir stick). - It is less on inspiration and more on expiration, so with airway constriction (asthma) "they're going to have just as much trouble moving the old air out … as getting new air in." - **Children** have smaller airways, so they get into trouble faster. - **Inspiratory wheezing is more worrisome than expiratory-only wheezing** (@1:30:34).

Gas exchange (Slides 91–108, @1:34:15–1:48:41)

Ventilation = air in and out of the alveoli. Perfusion = blood through the pulmonary capillaries. Diffusion = O2/CO2 across the alveolar-capillary membrane. (Her memory aid: a house's "ventilation system" moves air.)

Perfused but not ventilated (PF: alveolar collapse, airway obstruction, ↓ chest expansion, respiratory center depression): - **Nursing problem: Impaired gas exchange: "shunt"** (Slides 101–102). "the blood is being shunted from the venous circulation to the arterial circulation without having picked up oxygen first … we're basically wasting perfusion." Picture: "a big round black thing that's occluding the airway" (@1:42:50, re-transcribed).

Ventilated but not perfused: - **Nursing problem: Impaired gas exchange: "dead air space"** (Slides 103–104). "even though the air is there in the alveolus, it's basically dead … there's no blood coming through" (@1:44:16). - Example of disorder: **pulmonary embolism.** It

obstructs a branch of the pulmonary artery. The usual cause is a DVT that gets knocked loose, goes to the heart, and lodges in the lungs. - **Assessment findings** (@1:44:56–1:48:41): - **Poor gas exchange (V/Q mismatch) → hypoxia, shortness of breath, fast breathing, fast heart rate** - Possible lung ischemia → inflammation → **pleuritic chest pain, slight fever** - Severity depends on clot size. **Tiny clots may go unnoticed** (sometimes found only at autopsy). **Large clots → shock and death.** - **Interventions** (Slides 107–108, @1:47:30): - **Embolectomy** (surgical removal of the clot) - If the patient is not a surgical candidate: **thrombolytic therapy, followed by anticoagulants** to prevent recurrence. Thrombolytics may be less effective on a very large clot. - **Prevention:** on long flights, do stationary leg exercises (skeletal muscle pump) to prevent DVT.

Diffusion: - Partial pressure: gases move from higher to lower partial pressure, and "the bigger the gradient … the more gas is going to diffuse" (Slides 93–96). - Surface area: ↑ surface area → ↑ diffusion. - Thickness: ↑ thickness → ↓ diffusion. - Patho change: ↓ diffusion across the membrane. - PF: - ↓ partial pressure of gas - ↓ surface area (diseases that destroy the membrane, or lung resection) - ↑ thickness (swelling from infection, scar tissue) - **Nursing problem: Impaired gas exchange / poor oxygenation.** "the thicker the membrane, the less gas exchange, the worse the patient's oxygenation is going to be" (@1:40:53). - **Intervention she gives: supplemental O2** (mask or nasal cannula) "means the partial pressure of oxygen in the alveolus is going to be even greater … and so more oxygen is going to diffuse into the blood" (@1:38:28).

I. Gas transport (Slides 109–116, @1:48:41–1:57:00)

- **A. Oxygen:** about 1% is dissolved (PO2). This is "the only oxygen that can leave the blood, enter the cells". About 98–99% is on Hgb (SaO2), which acts as storage. Hgb normally starts releasing O2 when PO2 is about 60.
- a. Hgb affinity for O2 decreases (unloads sooner) when conditions reflect ↑ tissue metabolism and O2 need:
  - acidosis
  - CO2 ↑
  - heat ↑ (fever)
  - "acid, carbon dioxide, and heat are all byproducts of metabolism"
- b. Affinity increases when conditions reflect ↓ metabolism:
  - alkalosis
  - CO2 ↓
  - heat ↓
  - Hgb may hold O2 until PO2 is about 40
- **B. CO2:** 10% dissolved (PCO2), 30% on Hgb, 60% as bicarb. **$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons HCO_3^- + H^+$.** The body "can take a waste product, like carbon dioxide, and change it into something that it can actually use to keep you in acid-base balance."
- **C. Problems with gas exchange or transport:**
- **Hypoxia: ↓ oxygen supply to tissues** (Slides 117–122). This is **not the same as ischemia** (↓ blood flow, which also means ↓ glucose and waste removal). You can have hypoxia without ischemia, e.g., atelectasis.
  - **Assessment findings, grouped as SNS and CNS s/s:**
  - **SNS (fight or flight; appear fairly quickly): tachycardia, pale cool clammy skin, anxiety.**
  - **CNS (the brain depends on O2 for ATP): not thinking clearly → restless or agitated, acting differently than usual, impaired judgment → trouble with coordination, confusion or delirium → coma** (@1:57:30–1:59:50, re-transcribed).
  - **Mnemonic for early signs, "RAT"** (Slide 121): **R**estlessness, **A**gitation, **T**achycardia.
  - **Cyanosis** (Slide 122) **is a fairly LATE sign.** "We don't want to wait until our patient turns blue."
  - **Capillary refill checks circulation, NOT oxygenation.** "on the test, please don't tell me that you're going to assess somebody's oxygen status by checking their capillary refill" (@2:00:26).
- **Hypercapnia: ↑ CO2 in blood → CO2 narcosis** (Slide 123). "narco means sleep." The person gets progressively sleepy, then disoriented, then may fall into a coma. Its s/s resemble hypoxia's, because poor gas exchange usually causes both.
- **Respiratory failure** (Slides 124–125): the patient *is* breathing (this is not respiratory arrest), but the lungs fail to oxygenate or remove CO2.
  - Criteria: PO2 < 50 (normally > 80, up to 100) and/or PCO2 > 50 (normally about 35–45). These criteria are valid only for acute problems. Chronic patients may be alert with these numbers.
  - **"you have to look at your patient, don't just look at the numbers"** (@2:00:26–2:04:43).

II. Regulation of breathing (Slides 126–130, @2:04:43–end)

- **A. Respiratory center** (pons/medulla): very irregular breathing is a sign of brainstem injury.
- Automatic control uses input from chemoreceptors (medulla, carotid, aorta) and lung receptors.
- Voluntary control is an override from higher centers (cerebral cortex).
- **B. Autonomic nervous system** (controls airway diameter only):
- Sympathetic stimulation → **bronchodilation.** "if you're going to fight or run away, you have to have oxygen to do that."
- Parasympathetic stimulation → **bronchoconstriction.**

Not from the professor

(General nursing knowledge. She did not say these in the lecture or show them on the slides. Check them against the textbook before relying on them.) - ↑ **compliance:** the classic example of a disease causing loss of elastic recoil is **emphysema/COPD.** She described the mechanism but did not name the disease in the parts of the audio that could be recovered. Air-trapping (a vocab term) is the result. - She did not name a specific nursing diagnosis label for each row. The labels above ("Decreased cardiac output," "Impaired gas exchange," "Ineffective breathing / impaired airflow," "Risk for infection," "Acute pain") are standard NANDA-style wording built from her explanations. - Common bronchospasm interventions (bronchodilators) and common pneumothorax s/s (sudden dyspnea, absent breath sounds on the affected side, asymmetric chest rise) were not covered in this lecture.

Notes on sources

- About 1:04:30–1:11:00 of the main transcript was garbled. The bronchospasm and mucosal-swelling material (~1:06:20–1:07:40), the shunt passage (~1:42:45–1:44:30), and the hypoxia CNS signs (~1:58:40–2:00:00) were re-transcribed directly from the lecture audio (/tmp/c8.wav) for this guide. They are marked "re-transcribed," and their wording may differ slightly from her exact words.